

ATAR Testbeam Shared Trigger Prototype DAQ

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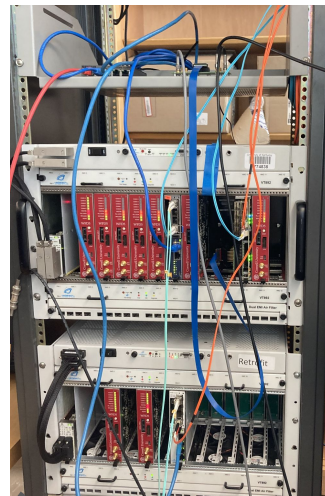
Current Teststand Setup

- The current hardware we have available in our UKy lab:
 - 1 HDSocV1 digitizer
 - 2 uTCA crates
 - ~10 working Cornell WFD5s
 - 1 FC7
 - 2 AMC13s
- Provided a shared external trigger to all front end electronics
- Rate tested up to ~150MB/s/~3kHz
 - Bottleneck is HDSoc's 55 MB/s limit
 - Event rate is “arbitrary”, but constrained by this



HDSocV1

+



g-2 style teststand

Run 244

Running

StopPause

Start: Thu Mar 12 16:27:08 2026

Alarms: On

Running time: 0h00m34s

runStatusSequencer

Data dir: /home/pioneer/data/gm2daq_data/

1773347234 16:27:14.798 2026/03/12 [mhttpd,INFO] Run #244 started

Equipment +

Status

Events

Events[/s]

Data[MB/s]

MasterGM2

MasterGM2@dhcp-10-163-105-238

2637

99.6

0.007

AMC13001

AMC13001@dhcp-10-163-105-238

2394

99.6

3.475

AMC13002

AMC13002@dhcp-10-163-105-238

2645

99.6

2.609

EB

Ebuilder@dhcp-10-163-105-238

2613

99.9

5.959

HDSoc-00

HDSoc00@::ffff:192.168.1.104

2992

99.6

2.056

Channel

Events

MB written

Compr.

Disk Level

#0: run00214.mid.gz4

0

0.000

0.0%

8.7%

Lazy Label

Progress

File Name

Files

Total

Clients

mserver [dhcp-10-163-105-238]

mhttpd [dhcp-10-163-105-238]

Logger [dhcp-10-163-105-238]

HDSoc00 [::ffff:192.168.1.104]

MasterGM2 [dhcp-10-163-105-238]

AMC13002 [dhcp-10-163-105-238]

AMC13001 [dhcp-10-163-105-238]

Ebuilder [dhcp-10-163-105-238]

Midas frontends creating data from shared trigger

Shared Trigger Scheme

- Each frontend electronic takes a shared external trigger
 - HDSocv1 frontend 0
 - AMC13 frontend 0
 - AMC13 frontend 1
- Trigger passes through FC7 for gating until run start
- Master frontend ungates only after all frontends complete begin-of-run
 - ensures readiness before triggers arrive
- Before other frontend end of end-of-runs, master frontend re-gates
 - Prevents data flow during other frontend end-of-run cleanups

Currently generated through a series of NIM electronics at a tunable rate

Generated Trigger Signal

Gates trigger signal until master frontend ungates

FC7

Fan out FC7 trigger from breakout board to HDSoc. AMC13s receive trigger from optical links.

FC7 Trigger

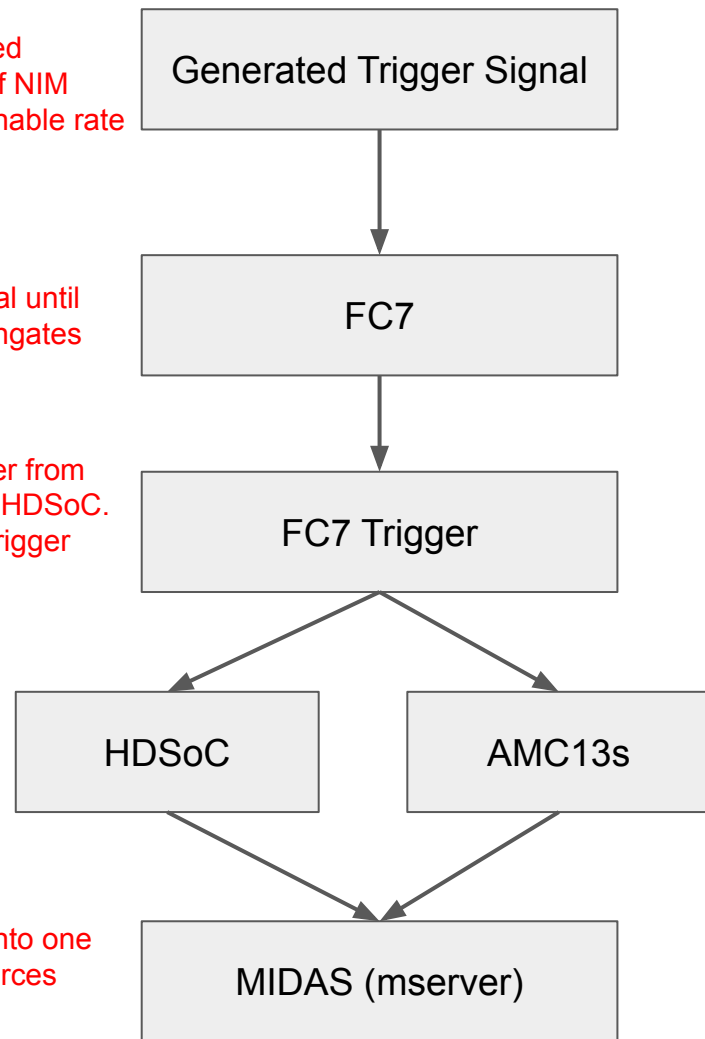
Frontend hardware process shared trigger.

HDSoc

AMC13s

Aggregates data into one shared event. Sources may be remote.

MIDAS (mserver)



TODO

- Provide more realistic “random” trigger
 - Current tests are a periodic configurable rate trigger
- Incorporate frontend simulators for SAMPic and MuPix
 - Don't have access to real hardware in our lab, but have simulator frontends available
 - SAMPic should (?) follow a similar shared trigger scheme as the HDSoc
 - Need to better understand MuPix trigger scheme to best design integration
- Put HDSoc on a shared clock
 - Should be possible, need guidance from Nalu/others
- HDSoc v2 integration (?)
- Incorporate new systems into DQM framework
 - SAMPic and HDSoc unpackers exist
 - HDSoc has plotting plugins for the webpage already
 - SAMPic does not have plotting plugins written, but this is a short exercise
 - Need to better understand MuPix data output to best design integration

Auxiliary Slides